Design and development Matrix

Needs analysis		Target specifications			Technical solution		Verification			Validation			Compliance
Ref Need	Description	Ref Spec	Description	Objectives measurable + tolerance	Possible solutions	The chosen one	Verification test description	Location/Team	Control methodology	Validation test description	Location/Team	Control methodology	Y/N
N1	Measure electrical signals generated by muscle activity	SP1.1	Acquire EMG signals that contain significant/measurable muscle activity	Set up hardware and record usable EMG signals during 4 daily activities			When recording EMG signals for four everyday activities, confirm measurable muscle activity has been recorded for each by inspecting raw recordings	2E Lab 2.44 / Group 27	Run tests on same individual and maintain electrode placement, sampling frequency and enviromental conditions. Record each test three times				
N2	Measure electrical activity from different muscle groups	SP2.1	Get a range of data from different muscle groups	Obtain recordings from biceps and forearm muscles			Do repeat testing to get an average for each type of muscle	2E Lab 2.44 / Group 27	Use the same individual for both muscle types				
N3	Acquire and store signal data	SP3.1	Have MATLAB store previous data	Use online EMG datasets to prototype our analysis pipeline in MATLAB			Import an EMG dataset, save it, reload the saved file and see if any data has been lost	2E Lab 2.44 / Group 27	Use the same dataset for this test				
N4	Minimise interference/noise aquired in EMG signals	SP4.1	Reduce baseline wandering	Butterworth filters (band-pass / notch) (~20 Hz – 500 Hz) Wavelet transform Moving average / RMS envelope Baseline offset / high-pass detrending			Apply filters and compare signal spectrum before and after to see if noise has been reduced	2E Lab 2.44 / Group 27	Use the same recorded EMG data, sampling frequency and spectral analysis settings for both comparisons. Keep filter parameters consistent				
		SP4.2	Reduce interference					2E Lab 2.44 / Group 27					
		SP4.3	Reduce motion artifacts					2E Lab 2.44 / Group 27					
		SP4.4	Minimise electrode contact noise					2E Lab 2.44 / Group 27					
		SP4.5	Remove high frequency electronic noise					2E Lab 2.44 / Group 27					
N5	Identify when signal quality is not sufficient for analysis	SP5.1	Detect excessive noise in the EMG recordings	Identify recordings where noise obscures the signal and prevents reliable signal extraction			Compare clean EMG singals with noisy ones to check if the system correctly identifies poor-quality signals	2E Lab 2.44 / Group 27	Use same signal, one with artificial noise added				
		SP5.2	Detect missing or incomplete data	Identify recordings with missing data points or incomplete measurements			Get system to analyse signals with missing data to see if it detects the fault	2E Lab 2.44 / Group 27	Use data where the location of data loss is known				
N6	Preserve important signal data	SP6.1	Ensure filtering did not remove key signal data	Check by comapring filtered signal to EMG database			Compare raw and filtered EMG recordings to assess whether important signal characteristics are preserved after filtering	2E Lab 2.44 / Group 27	Use the same original EMG dataset for both comparisons keeping the sampling frequency consistent				
N7	Identify when muscles are in contraction and relaxation	SP7.1	Produce simple classification between resting and contracted state	Compare activities to a baseline graph for when muscles are at rest			To get an accurate reading of rest to compare to, ensure the participant is still with their arm resting on their leg and repeat this three times to obtain an average	2E Lab 2.44 / Group 27	Ensure the same participant that does all the activities is also the one whose baseline rest is compared to				
N8	Compare muscle activation across different activities	SP8.1	Look at muscle activation when writing	Retrieve data when writing one line on standard A4 paper (X3)			Compare EMG signals when writing to when at rest to confirm muscle contraction is being picked up	2E Lab 2.44 / Group 27	Keep the participant, sentence, pen, writing posture and electrode placement consistent				
		SP8.2	Look at muscle activation when picking up a mug	Retrieve data when lifting a mug from a table height to mouth (X3)			Compare EMG signals when lifting mug to when at rest to confirm muscle contraction is being picked up	2E Lab 2.44 / Group 27	Keep the participant, mug, posture and electrode placement consistent				
		SP8.3	Look at muscle activation when at rest	Retrieve data when arm is resting on leg for 10s interval (x3)			Compare data taken at rest with signal of muscle contraction to verify lower muscle activation has been detected	2E Lab 2.44 / Group 27	Keep the participant, arm position, electrode placement and recording duration (10 seconds) consistent				
		SP8.4	Look at muscle activation when clenching fist	Retrieve data when fist is clenched for 5s (x3)			Compare EMG signals when clenching first to when at rest to confirm muscle contraction is being picked up	2E Lab 2.44 / Group 27	Keep the participant, posture and electrode placement consistent				
		SP8.5	Look at muscle activation when opening a door	Retrieve data when opening a door from fully closed to fully open (x3)			Compare EMG signals when opening door to when at rest to confirm muscle contraction is being picked up	2E Lab 2.44 / Group 27	Keep the participant, door, posture and electrode placement consistent				
N9	Detect changes in EMG signals associated with muscle fatigue over a period of time	SP9.1	Detect features that indicate muscle fatigue which shifts the spectrum to lower frequencies with sustained muscle contraction	Calculate mean / median frequency of the power spectrum			Analyse EMG data with known frequency changes and check whether the system correctly detects them	2E Lab 2.44 / Group 27	Use the same EMG data, sampling frequency and analysis settings for all tests				
N10	Detect key features of the EMG signal	SP10.1	Detect signal envelope from a sliding time window	Use 100-250 ms for amplitude-domain features and 500-1000 ms for frequency domain features			Analyse a known EMG dataset and compare the calculated signal envelope, RMS and MAV values against reference results to check accuracy	2E Lab 2.44 / Group 27	Use the same EMG dataset, sampling frequency and window settings for all calculations				
		SP10.2	Calculate root mean squared (RMS) amplitude					2E Lab 2.44 / Group 27					
		SP10.3	Calculate mean absolute value (MAV)					2E Lab 2.44 / Group 27					
N11	Produce reliable and repeatable measurements in controlled conditions	SP11.1	Keep conditions controlled	Use the same individual for measurements and the same EMG device to keep controlled conditions			Results should be consistent under the same enviornment and test to show that the eviornment is controlled	2E Lab 2.44 / Group 27	Nominate a memeber at the start and ensure they are present for all the tests				
		SP11.2	Do repeat tests	Repeat all actions three times			All results should be withtin a reasonable uncertainty range	2E Lab 2.44 / Group 27	Calculate a mean and an uncertainty for each set of values to determine what defines an anomaly				

N12	Produce visual representation of data	SP12.1	Convert raw data to graphical visuals to aid key feature analysis	Produce line graphs in MATLAB with legends, titles and units			Produce line graphs and compare them to see if they are consistent in style to ensure no key features are missing	2E Lab 2.44 / Group 27	Produce all graphs on the same software (ie MATLAB)				
N13	Safe and non-invasive	SP13.1	Ensure electrical safety during operation	Follow manufacture and university guidance			Compare the procedure with the completed risk assessment	2E Lab 2.44 / Group 27	Make risk assessment before conducting tests				
		SP13.2	Minimise test person discomfort	Report any skin ieration and stop testing imediately			Be gentle when attaching probes	2E Lab 2.44 / Group 27	Make test subject aware of any risks				
N14	Be made with supplied lab equipment	SP14.1	Only use components available in the EMG kit supplied by the lab	Collect the EMG kit from the lab promptly as kits are allocated on a first comes first served basis			Compare two of the same EMG kits to ensure no equipment or materials are missing	2E Lab 2.44 / Group 27	Compare with component checklist				
N15	Have results in a format understandable for a non-technical audience	SP15.1	Explain findings in accessible language	Explain key techniques and terms simply using clear language and real world examples to relate back to			Do a mock presentation with a non-technical audience to receive feedback on whether they understood the terminology used	2E Lab 2.44 / Group 27	Ask each audience member to give their feedback anonymously and individually to remove any bias				
		SP15.2	Present EMG results using clear visuals	All graphs must have clear legends, axes and titles			Do a mock presentation with a non-technical audience to receive feedback on whether they understood the graphics used	2E Lab 2.44 / Group 27					